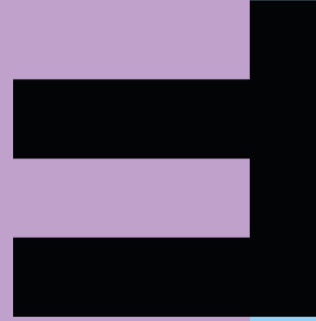


FHV

Vorarlberg University
of Applied Sciences



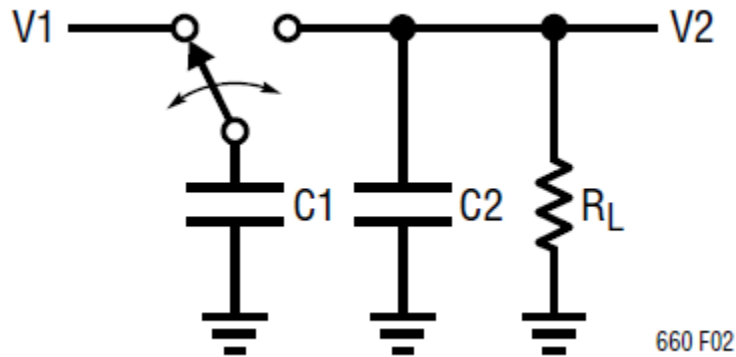
Analog Stuff

15.04.2026

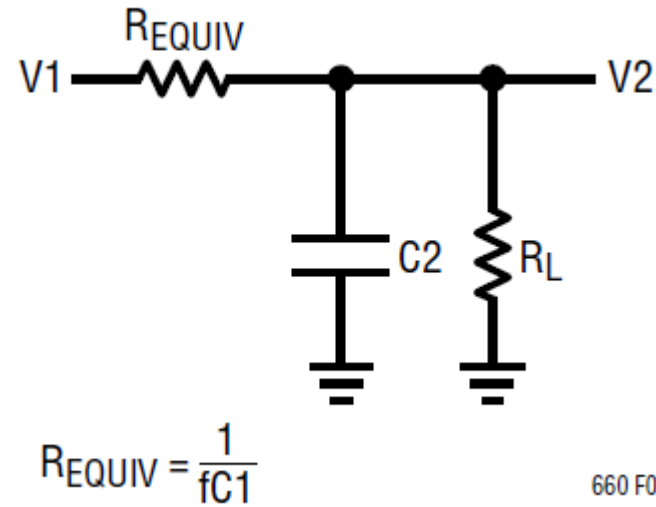
EIT & MEC Bachelor

Horatiu O. Pilsan

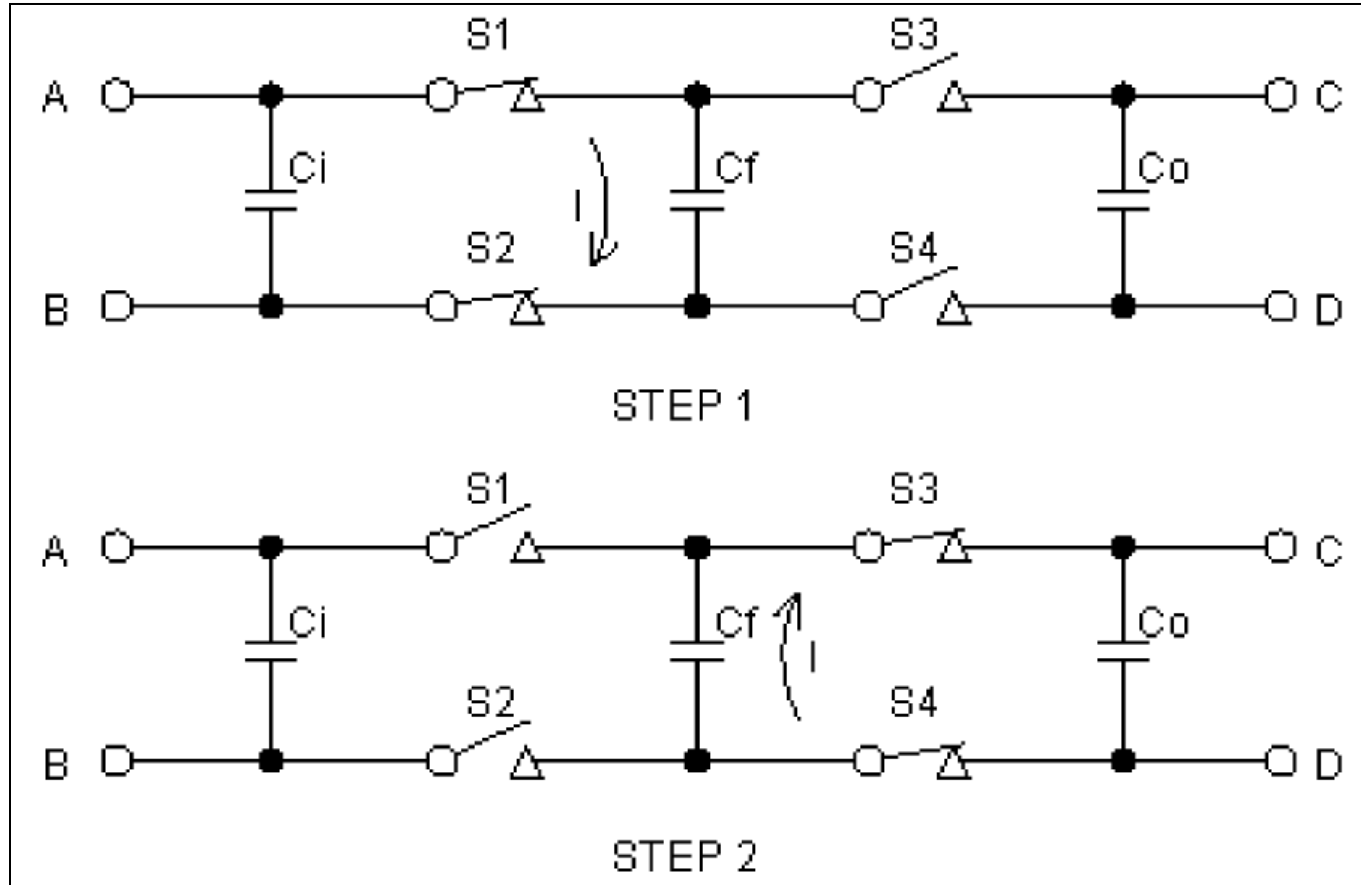
Charge Pump Principle



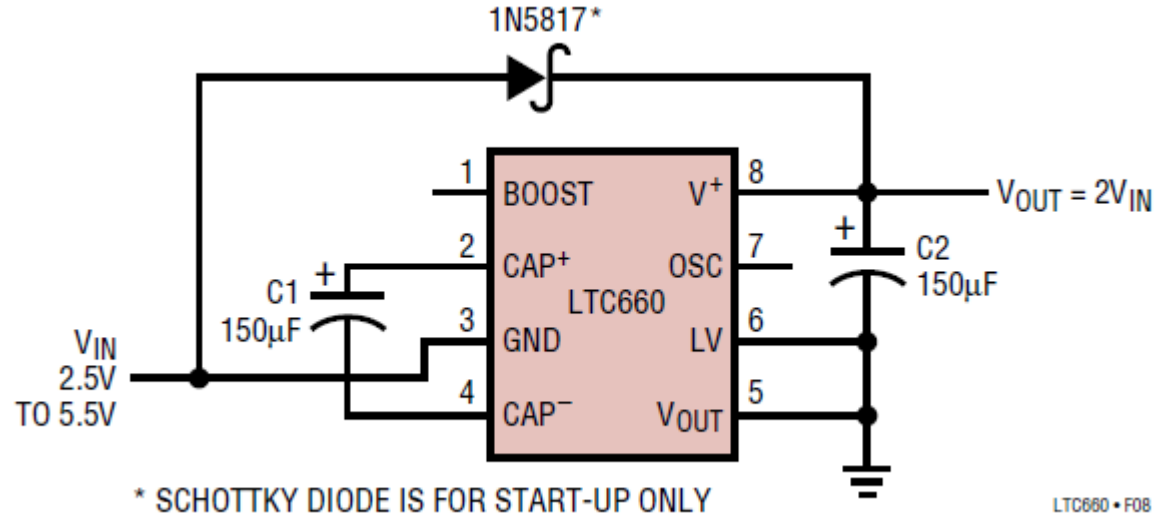
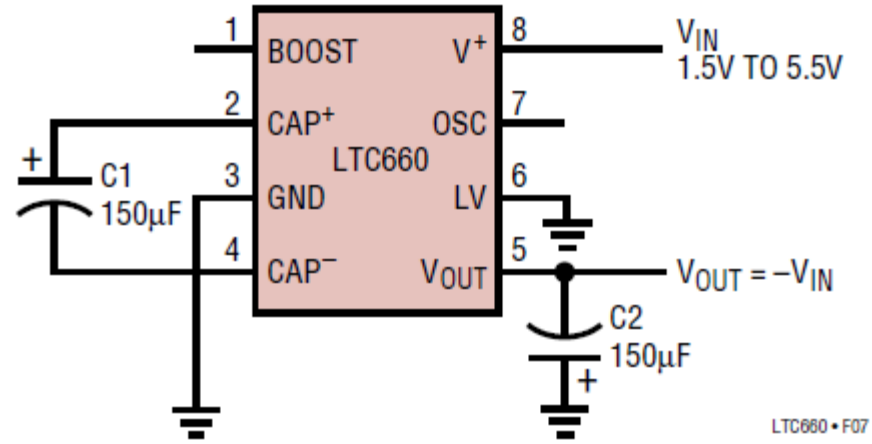
[5]



Charge Pump Details



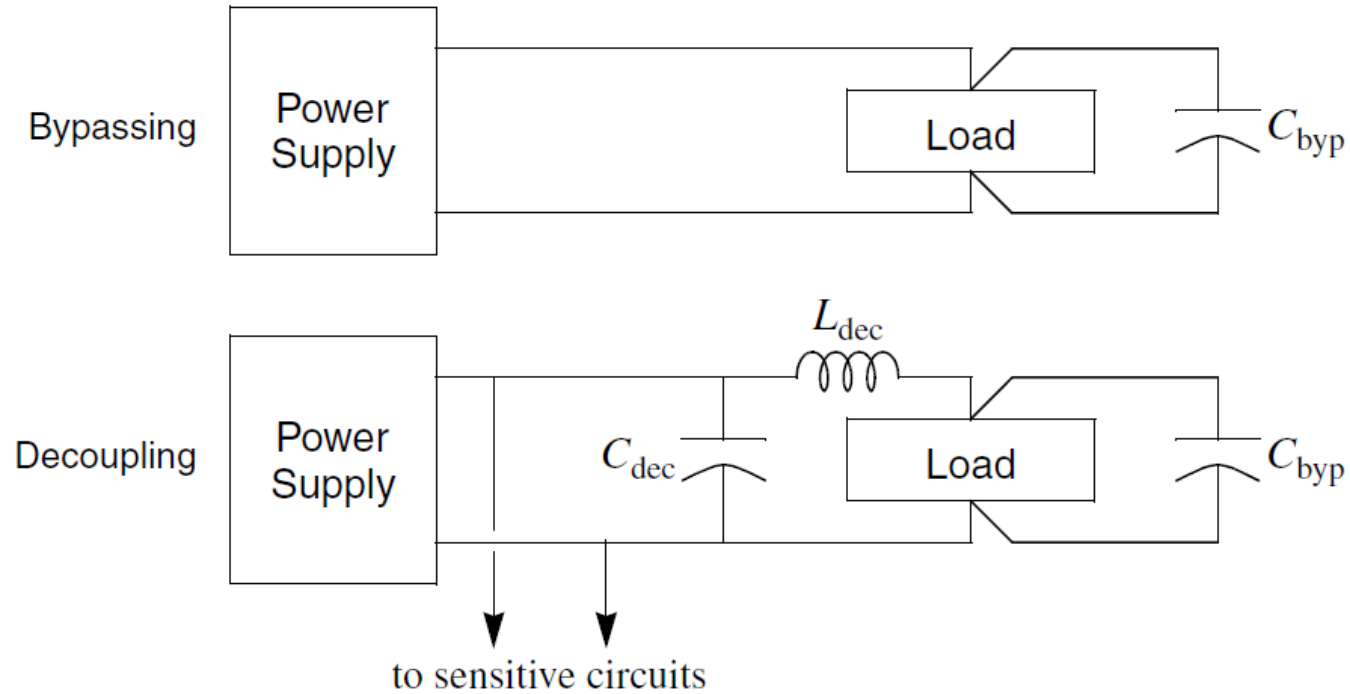
Charge Pump Application



[5]



Bypassing - Decoupling



[6]



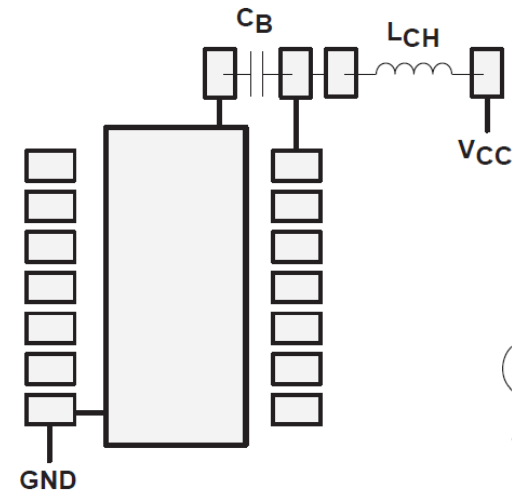
Decoupling an Analog IC

- Why do we need to decouple analog ICs?
- Is this bypassing or decoupling?
- How is this done?
- Which information on the IC's sensitivity can be found in data sheet?
- See: [7] - a good reference on decoupling



Traditional Bypassing a Digital IC

- Main problem: inductivity of the power supply lines (and of the ground line obviously too)
- Classical method: each IC gets a bypassing capacitor (e.g. 100 nF), must-be ceramic type, best suited is a SMD!



[8]

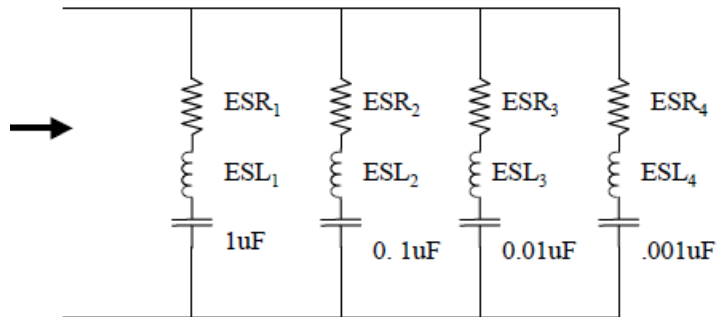
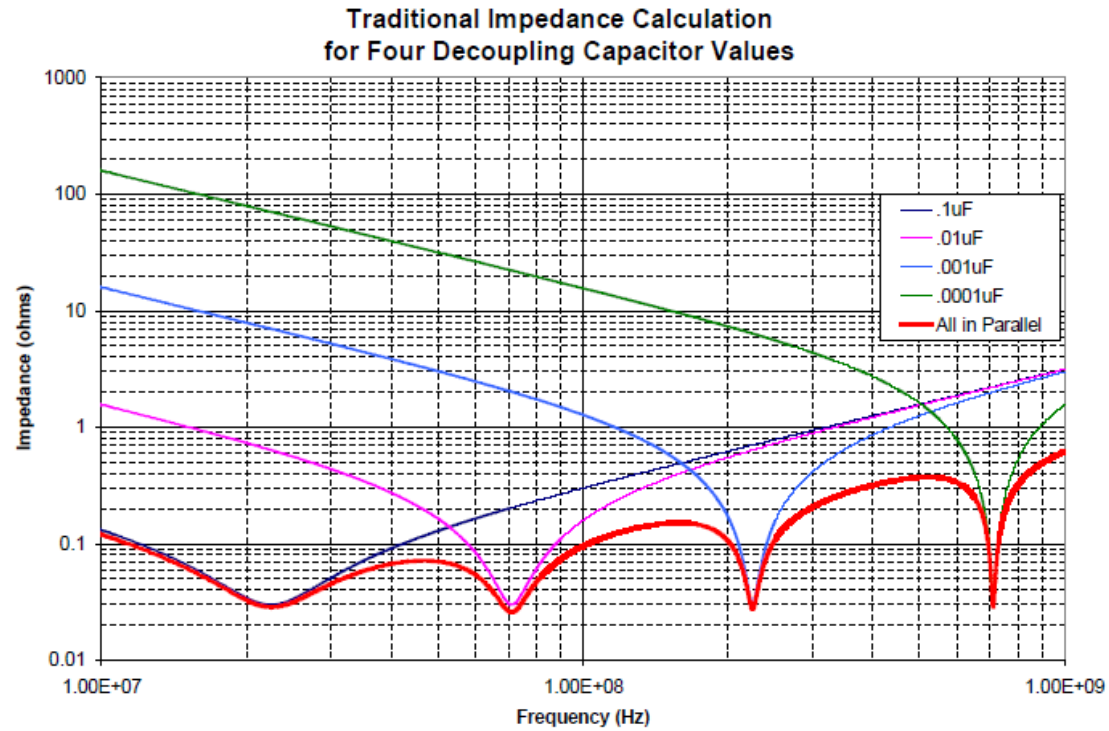


Modern Bypassing

- But capacitors differ in behavior,
- Frequency increases and signal edges get faster,
- And single sided or two sided PCB are no more in use.
- => Bypassing is a layout issue!

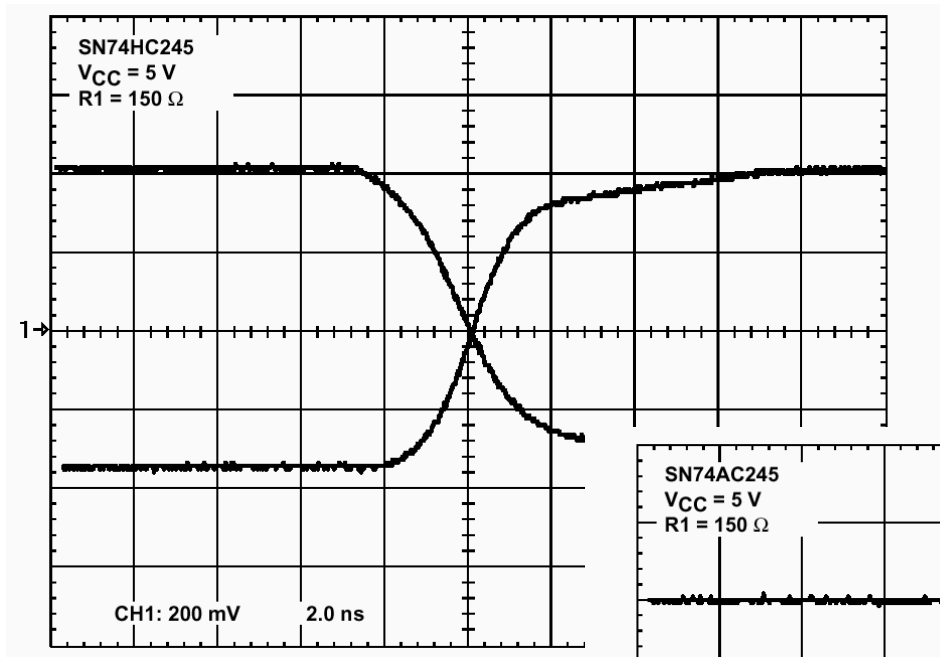


Capacitors

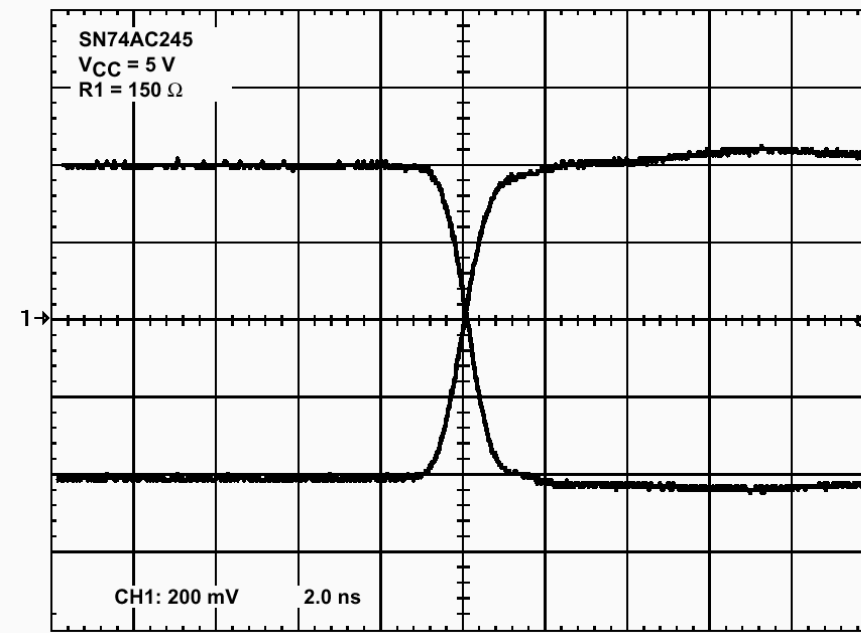


[4]

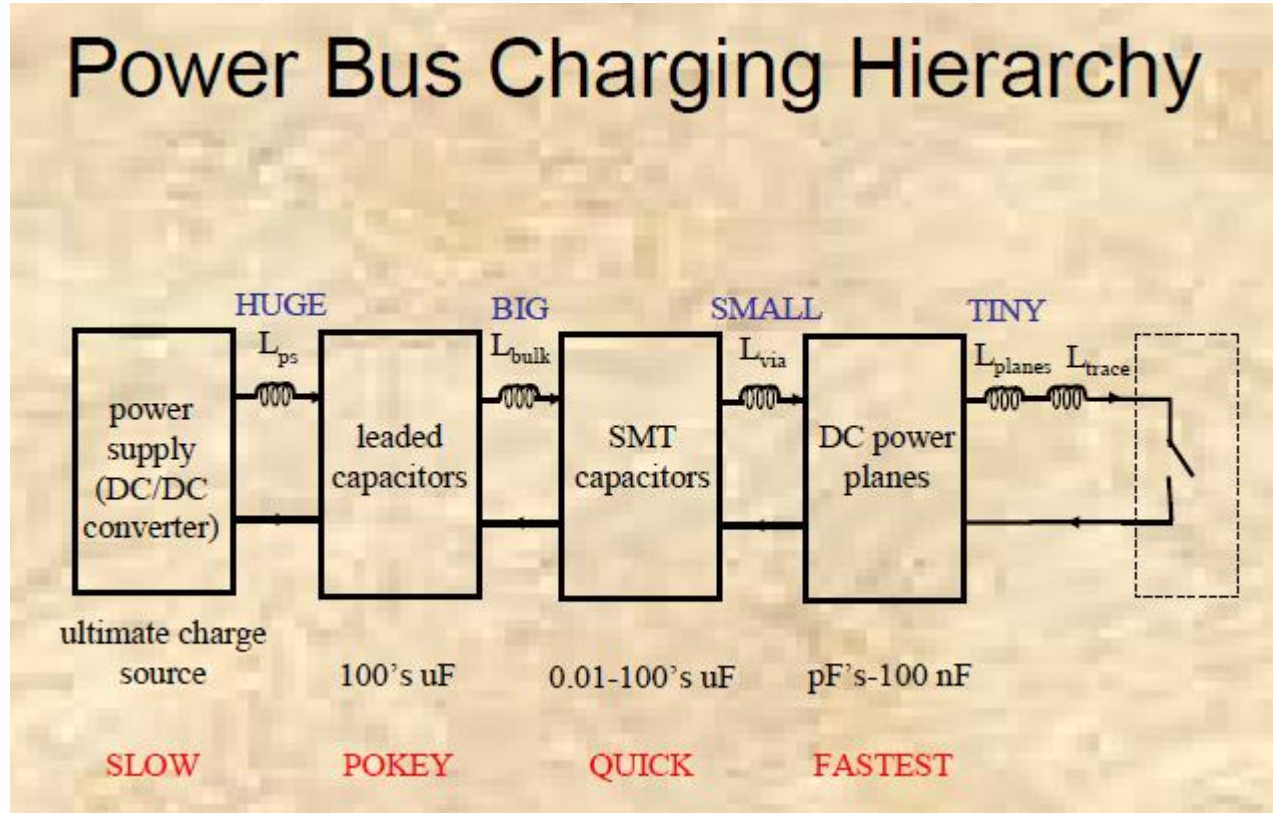
Signal Edges



[9]



Multilayer Bypassing



References

- [1] Tietze, Ulrich; Schenk, Christoph (1999): Halbleiterschaltungstechnik. 11. Auflage. Berlin: Springer
- [2] Hering, Eckbert; Bressler, Klaus; Gutekunst, Jürgen (2014): Elektronik für Ingenieure und Naturwissenschaftler, 6. Auflage, Berlin, Heidelberg: Springer
- [3] „TLC2932 Application Report”, Texas Instruments, 1997, document SLLA011B, <http://www.ti.com/lit/an/slaa011b/slaa011b.pdf>
- [4] Clock Driver Circuit Overview, www.ti.com
- [5] LTC660 Data Sheet, Linear Technology, <http://cds.linear.com/docs/en/datasheet/660fa.pdf>
- [6] „Power Supply Noise Reduction”, Ken Kundert, Designer’s Guide Consulting, Inc. <http://www.designers-guide.org/Design/bypassing.pdf>
- [7] MT 101 Tutorial, Analog Devices, <https://www.analog.com/media/en/training-seminars/tutorials/MT-101.pdf>
- [8] „Printed-Circuit-Board Layout for Improved Electromagnetic Compatibility”, Texas Instruments, document SDYA011, <http://www.ti.com/lit/an/sdya011/sdya011.pdf>
- [9] „Electromagnetic Emissions From Logic Circuits, Application Report”, Texas Instruments, document SZZA007, <http://www.ti.com/lit/an/szza007/szza007.pdf>



Analog Stuff

15.04.2026

EIT & MEC Bachelor

Horatiu O. Pilsan

